

LG Soft India — All Programming Questions

Confidential — Internal Use Only

Total programming questions: 8

1. Count and Say

Given a positive integer n , return the n th term of the count-and-say sequence. The sequence starts with "1", and each next term is formed by reading off the digits of the previous term, counting each run of consecutive identical digits and stating the count followed by the digit. For example, "1" is read as "one 1" and becomes "11"; "11" is read as "two 1s" and becomes "21"; "21" is read as "one 2, one 1" and becomes "1211".

Input: $n = 4 \rightarrow$ Output: "1211"

Link: leetcode.com/problems/count-and-say/

Source: LeetCode #38

2. Absolute Pair Sum Closest to Zero

Given an array of integers containing both positive and negative values, find the pair of elements whose sum is closest to zero. If two pairs are equally close, the pair with the larger sum is preferred.

Input: $arr = [-8, 5, 2, -6] \rightarrow$ Output: -1 (pair 5, -6)

Link: [geeksforgeeks.org/find-pair-sum-closest-0/](https://www.geeksforgeeks.org/find-pair-sum-closest-0/)

Source: GeeksforGeeks

3. Isomorphic Strings (Couple Strings)

Given two strings, determine whether the characters of the first string can be mapped one-to-one onto the characters of the second string such that the mapping is consistent across the entire string — no two characters map to the same character, and the same character always maps to the same character.

Input: "paper", "title" $\rightarrow$ Output: YES ($p \rightarrow t, a \rightarrow i, p \rightarrow t, e \rightarrow l, r \rightarrow e$ — consistent mapping)

Link: leetcode.com/problems/isomorphic-strings/

Source: LeetCode #205

4. Missing Element (Array N and N-1)

Given two arrays of integers — the first with N elements and the second with $N-1$ elements formed by removing exactly one element from the first — find and return the missing element.

Input: $Array1 = [2, 1, 2, 1]$, $Array2 = [1] \rightarrow$ Output: 2

Link: [geeksforgeeks.org/find-missing-number/](https://www.geeksforgeeks.org/find-missing-number/)

Source: GeeksforGeeks

5. Longest Consecutive Sequence

Given an unsorted array of integers, find the length of the longest run of consecutive integers, without requiring the elements to be consecutive in the array itself. The solution should run in $O(n)$ time.

Input: $[100, 4, 200, 1, 3, 2] \rightarrow$ Output: 4 (sequence 1, 2, 3, 4)

Link: leetcode.com/problems/longest-consecutive-sequence/

Source: LeetCode #128

6. Backspace String Compare

Given two strings that may contain the '#' character (representing a backspace that deletes the previous character), determine whether the two strings are equal once all backspaces have been applied.

Input: $s = "ab\#c"$, $t = "ad\#c"$ $\rightarrow$ Output: true (both become "ac")

Link: leetcode.com/problems/backspace-string-compare/

Source: LeetCode #844

7. Card Shuffle (closest match: Reorganize String)

Given a string, rearrange its characters so that no two adjacent characters are the same. Return any valid rearrangement, or an empty string if it isn't possible.

Input: "aab" $\rightarrow$ Output: "aba"

Link: leetcode.com/problems/reorganize-string/

Source: LeetCode #767

8. Card Shuffle (same greedy logic: Task Scheduler)

Given a list of tasks and a cooldown period n , find the minimum number of time units the CPU needs to complete all tasks, given that the same task must be separated by at least n intervals (idle slots allowed).

Input: tasks = [A,A,A,B,B,B], $n = 2 \rightarrow$ Output: 8

Link: leetcode.com/problems/task-scheduler/

Source: LeetCode #621